

UCS415:DESIGN AND ANALYSIS OF ALGORITHMS

L	T	P	Cr
3	0	2	4.0

Course Objective: To provide students with the knowledge and skills necessary to design and analyse algorithms for solving computational problems.

Syllabus:

Introduction and Complexity Analysis: Analysing algorithms, Complexity classes, Time and space trade-offs in algorithms, Recurrence relations, Analysis of iterative and recursive algorithms, Amortized Analysis.

Algorithm Design Techniques and Analysis

Divide and Conquer: Fundamentals of divide and conquer strategy, Applications such as The maximum subarray problem, Strassen's algorithm for matrix multiplication, merge sort, quick sort etc.

Greedy Algorithms: Elements of greedy strategy, Sustainable resource allocation using greedy algorithms, Applications such as activity selection, Huffman Coding for energy efficiency in data transmission, job sequencing, fractional knapsack problem, waste bin placement, load balancing in smart grid etc.

Dynamic Programming: Elements of dynamic programming, Memorization and tabulation approaches, Applications such as matrix multiplication, 0/1 knapsack (Optimizing Energy Storage), Longest common subsequence, Optimal binary search tree, etc.

Backtracking: Introduction, Applications such as N queen problem, sum of subsets, graph coloring, etc.

Branch and Bound Algorithm: General method, Applications such as 0/1 knapsack problem, Traveling salesperson problem etc.

Graphs & Algorithms: Introduction to graphs, Paths and Circuits, Euler Graphs, Hamiltonian graphs, Applications of Eulerian/Hamiltonian Graph in urban planning, Cut-sets, Strongly connected component, Topological sort, Max flow: Ford Fulkerson algorithm, max flow- min cut, Application of Ford Fulkerson in smart grid energy distribution.

String Matching Algorithms: Suffix arrays, Rabin-Karp, Knuth-Morris Pratt (KMP), Boyer Moore algorithm.

Problem Classes: P, NP, NP-Hard and NP-complete, deterministic and non-deterministic polynomial time algorithm approximation, Randomized algorithms.

Laboratory Work (if applicable): Implementation of various algorithmic techniques for solving common computational/engineering problems.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyse the complexity of algorithms and implement it in a specific scenario.
2. Apply common algorithmic techniques such as greedy, dynamic programming etc. to standard computational problems.
3. Design solutions using appropriate data structures and algorithms, such as string matching, randomized, approximation and graph algorithms.
4. Develop efficient algorithms to solve various computationally complex problems in computing.

Text Books

1. Cormen H. T., Leiserson E. C., Rivest L. R., and Stein C., Introduction to Algorithms, MIT Press (2009) 3rd ed.
2. Horwitz E., Sahni S., Rajasekaran S., Fundamentals of Computers Algorithms, Universities Press (2008) 2nd ed.

Reference Books

1. Levitin A., Introduction to the design and analysis of algorithms, Pearson Education (2008) 2nd ed.
2. Aho A.V., Hopcraft J. E., Dulman J. D., The Design and Analysis of Computer Algorithms, Addison Wesley (1974) 1st ed.
3. Sedgewick R. and Wayne K., Algorithms, Addison-Wesley Professional (2011), 4th ed.